

Homework 03: Chapter 04 and 06

EEC 3150

For each Python exercise, write a separate Python script. Submit all scripts on Blackboard under the appropriate assignment. Name your script files with the following format:

HW<assignment number>_Ex<exercise number>_<FirstInitial><LastName>.py

Example: HW01_Ex01_GWest.py

For short answer or math exercises, either do your work on a sheet of paper and upload a picture of it to Blackboard or submit some kind of text file or Word document to Blackboard. You can place all such exercises in a single document as long as you clearly label each exercise. These files should follow a similar naming convention as the Python scripts.

Exercise 1 (10 points, Python)

Write a function `Reverse(s)` which will accept a single string argument and return the reversed string. Call the function. Here's an example of how to use the function:

```
>>> Reverse('abcdefg')
'gfedcba'
```

DO NOT use any built-in string methods.

Exercise 2 (10 points, Python)

Write a function `Split(s, n)` which will accept two arguments (a string and an integer n) and return two values (the left and right halves of the string when split at the n -th index of the string). Call the function. Here's an example of how to use the function:

```
>>> Split('abcd', 2)
('ab', 'cd')
```

DO NOT use any built-in string methods.

Exercise 3 (10 points, Python)

Write a function `ConcatStrings(*s)` which will accept a variable number of string arguments and return the concatenation of all of them. Demonstrate the function by calling it. Here's an example of how to use the function:

```
>>> ConcatStrings('a', 'b', 'c')
'abc'
```

Exercise 4 (10 points, Python)

Write a function `trig(x, n)` which will accept two arguments (a float x and an integer n) and return two values: the n -th order Taylor approximation of $\sin(x)$ and $\cos(x)$.

$$\sin(x) \approx \sum_{i=0}^n \frac{(-1)^i x^{2i+1}}{(2i+1)!}$$
$$\cos(x) \approx \sum_{i=0}^n \frac{(-1)^i x^{2i}}{(2i)!}$$

Test the function by plugging in an approximation for π for x . You may use my factorial code or write your own.

Exercise 5 (20 points, Python)

Write a program with two functions—one non-recursive (`HarmNon(n)`) and one recursive (`HarmRec(n)`)—that will calculate the Harmonic series:

$$\sum_{i=1}^n \frac{1}{i} = \frac{1}{1} + \frac{1}{2} + \cdots + \frac{1}{n}$$

Test your function by printing the values of both functions with something like the following code:

```
n = 10
for i in range(1, n+1):
    print(HarmNon(i), HarmRec(i))
```